OffBytes Lab

Prepared for:

Looputo Finance

October 30, 2026

Looputo Finance Audit Report

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1. Executive Summary

About Looputo

This audit focuses on Looputo Finance's leveraged-position pools, which let users deposit assets into a shared pool and take on leverage of their choosing while keepers monitor the pool to limit liquidation and negative-carry risk. Each pool tracks per-user collateral and debt shares and sources its collateral supply and debt borrowing from an Aave V3, Spark or Morpho Blue lending adapter, with position changes executed atomically through Morpho flash loans and swaps on owner-approved DEX aggregators. Pools are deployed as beacon proxies from an owner-controlled factory that fixes the assets, lending market, risk limits, fees, aggregators and keeper at initialization.

Risk Profile

The following table summarizes the distribution of identified vulnerabilities by risk level:

Risk Level	Count	Fixed	Acknowledged
Critical	0	0	0
High	0	0	0
Medium	0	0	0
Low	1	1	0

Audit Scope

- Repository: https://github.com/looputo-finance/Looputo-Finance
- Audit start date: October 23, 2026
- Commit hash: 1109e7303674504768399b66e34d3865d1077a18
- Files/Modules in Scope:
  - `src/LoopPool.sol`
  - `src/MorphoLoopPool.sol`
  - `src/AaveV3LoopPool.sol`
  - `src/SparkLoopPool.sol`
  - `src/Factory.sol`
  - `src/LoopPoolBeacon.sol`
  - `src/libraries/SharesLib.sol`
  - `src/libraries/MathLib.sol`

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2. Assumptions and Considerations

Audit Assumptions

- I. Morpho Blue, Aave V3 and Spark core contracts are treated as trusted dependencies and behave as documented.
- II. DEX aggregators approved by the owner return the swapped assets to the pool and are not themselves malicious; swap calldata is supplied by the user and validated only by the pool's slippage checks.
- III. The collateral and debt tokens configured for a pool are standard ERC-20 tokens without fee-on-transfer or rebasing behaviour.

Centralization Risks

- IV. The `Factory` owner is a trusted, protocol-controlled address (governance multisig). It can change LTV limits, caps, fees, approved aggregators, the keeper, and upgrade every pool through the beacon.
- V. The keeper is a trusted, protocol-controlled operator that may batch `deleverage` / `releverage` user positions within the configured LTV bounds.
- VI. The fee collector is a passive protocol-controlled address.

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3. Severity Definitions

Each issue identified in this report is assigned a severity level based on two dimensions: Impact and Likelihood. These dimensions help project our team's understanding of both the potential consequences of a vulnerability and how likely a vulnerability is to be discovered and exploited in the real world.

Impact

Impact reflects the potential consequences of the issue—particularly on project funds, user funds, and the availability or integrity of the protocol.

- **High Impact:** Successful exploitation could result in a complete loss of user or protocol funds, disruption of core protocol functionality, or permanent loss of control over critical components.
- **Medium Impact:** Exploitation could cause significant disruption or partial loss of funds, but not a total compromise. May impact some users or non-core functionality.
- **Low Impact:** The issue has minor or negligible consequences. It may affect edge cases, expose metadata, or degrade performance slightly without putting funds or core logic at serious risk.

Likelihood

Likelihood reflects how easy a vulnerability is to discover and exploit by an attacker, as well as how economically attractive the exploit is to an attacker.

- **High Likelihood:** The vulnerability is trivially exploitable. It can be exploited by a wide range of actors without privileged access rights, with minimal capital requirements and low financial risk.
- **Medium Likelihood:** The vulnerability can be found and exploited with moderate effort. It might require a significant capital investment, but with manageable financial risk.
- **Low Likelihood:** Exploitation is often technically unfeasible or requires highly specialized conditions. It may require extraordinary effort or significant financial risk for an attacker, with a high chance of failure and minimal potential return.

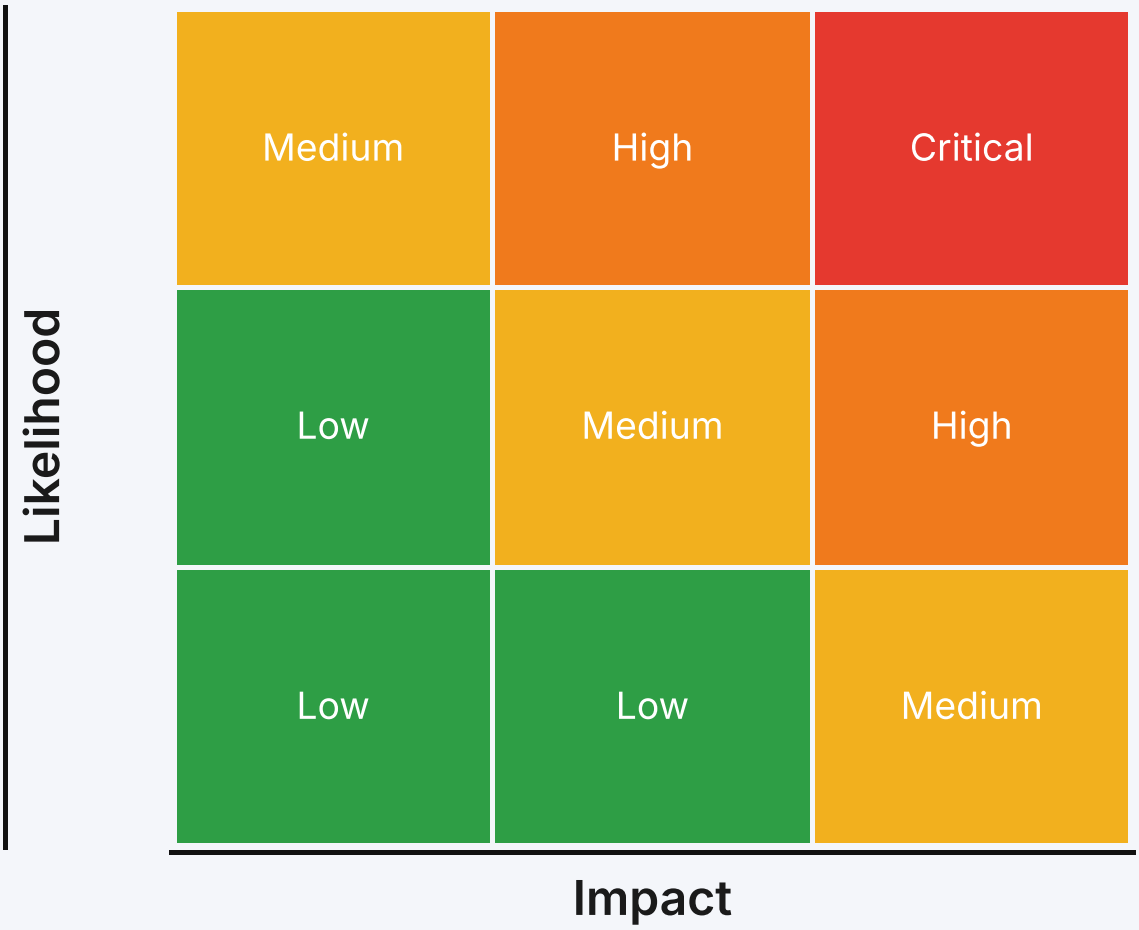
Severity Classification Matrix

By combining Impact and Likelihood, we assign a severity level using the matrix below:

- **Critical:** High impact + high likelihood (e.g. a bug that could allow anyone to drain a substantial amount of protocol funds with minimal effort)
- **High:** High impact with medium likelihood, or medium impact with high likelihood
- **Medium:** Moderate impact and/or discoverability
- **Low:** Minimal impact or unlikely to be exploited

This structured approach helps teams prioritize fixes and mitigate the most dangerous threats first.

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Severity Matrix

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4. Findings

L01: Keeper rebalances leave equity cap stale

Status: Resolved

Impact: Low

Likelihood: Medium

Severity: Low

Location: https://github.com/looputo-finance/Looputo-Finance/blob/1109e73/src/LoopPool.sol#L690-L802

>_ LoopPool.sol SOLIDITY

```
1123:     function _increaseNetEquity(bool isColl, uint256 amount) internal {
1124:         if (isColl) {
1125:             uint256 equivalentAmount = convertCollToDebt(amount);
1126:             poolNetEquity += equivalentAmount;
1127:         } else {
1128:             poolNetEquity += amount;
1129:         }
1130:
1131:         require(poolNetEquity <= cap, "Deposit amount exceeds cap");
1132:     }
1133:
1134:     function _decreaseNetEquity(uint256 amount) internal {
1135:         if (amount > poolNetEquity) {
1136:             poolNetEquity = 0;
1137:         } else {
1138:             poolNetEquity -= amount;
1139:         }
1140:     }
```

Description:

`poolNetEquity` is the value the deposit cap is enforced against. It is increased only on user deposits (`openPosition`, `addPositionEquity`) and decreased only on user withdrawals (`reducePositionEquity`, `closePosition`).

The keeper-only `deleverage` and `releverage` flows change the pool's collateral and debt balances in the lending market and mint or burn shares accordingly, but never touch `poolNetEquity`. Because the swap leg in `onMorphoFlashLoan` is bounded only by the flash-loan repayment amount, a keeper rebalance can realise a swap loss or gain that is reflected in the pool's real equity but not in the stored value.

A loss-making unwind therefore leaves the cap's equity accounting overstated and can block otherwise valid deposits, while a gain-making rebalance leaves it understated and permits deposits beyond the configured cap. The only correction is the owner manually overwriting the value through `syncPoolNetEquity`.

Recommendation:

After each keeper rebalance, reconcile `poolNetEquity` with the post-rebalance debt-denominated collateral value less debt, so subsequent cap checks use the pool's actual equity rather than a stale stored figure.

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DIFF

```
        _checkSystemLTV();
+       _reconcileKeeperEquity(
+           cache.totalCollAmountAfter,
+           cache.totalDebtAmountAfter
+       );
+
        _distributeDeleveragedShares(params, cache);
        ...
        _checkSystemLTV();
+       _reconcileKeeperEquity(
+           cache.totalCollAmountAfter,
+           cache.totalDebtAmountAfter
+       );
+
        _distributeReleverageShares(params, cache);
        ...
+   function _reconcileKeeperEquity(
+       uint256 collAmount,
+       uint256 debtAmount
+   ) internal {
+       uint256 collValue = convertCollToDebt(collAmount);
+       poolNetEquity = collValue > debtAmount ? collValue - debtAmount : 0;
+   }
+
    function _increaseNetEquity(bool isColl, uint256 amount) internal {
```

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5. About Us

About OffBytes Lab

OffBytes Lab is an independent blockchain security firm focused on smart-contract audits.

Our researchers come from competitive audit platforms such as Code4rena and Sherlock and have a track record of validated High- and Medium-severity findings across DeFi, bridge, staking and NFT protocols.

Every engagement combines rigorous manual review with executable proof-of-concept tests, so that each reported issue is demonstrated rather than theorised and each fix is re-verified against the same tests.

Audit Methodology

Our audit methodology is specialised to provide thorough security assessments of EVM smart contracts. We focus on rigorous manual analysis, detailed threat modelling, executable proof-of-concept validation, and careful review of implemented fixes.

1. Program Context and Architecture Analysis

We begin by studying the protocol's documentation and intended functionality, mapping contract interactions, transaction flows and state management. Special attention is given to how the protocol interfaces with common EVM standards such as ERC-20, ERC-721 and ERC-4626, with proxy and governance modules, and with external protocols, verifying that their inputs and return values are handled correctly.

2. Threat Modelling

We define attacker capabilities for the EVM execution environment and enumerate the vulnerability classes relevant to the codebase, including but not limited to:

- Improper access control, role checks or `msg.sender` verification
- Unauthorised state-variable modification
- Misuse of low-level calls, `delegatecall`, or proxy / upgradeability patterns (storage collisions, initialisation)
- Incorrect handling of external calls, including reentrancy vectors
- Denial-of-service vectors related to gas limitations or reverting dependencies

3. In-depth Manual Security Review

Our auditors perform a comprehensive line-by-line inspection of the source code, focusing on:

- Missing or insufficient access control and role checks
- Inadequate `msg.sender` and transaction-origin checks
- Incorrect handling of external calls and callback privileges, including reentrancy
- Arithmetic overflow / underflow and rounding errors
- Unsafe use of ABI encoding / decoding or low-level assembly
- Improper ERC-20 / token handling (approval logic, flash-loan vectors, token locking)
- Logic flaws in financial operations or state transitions

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- Edge-case handling in function input validation
- Denial-of-service vectors related to gas usage and transaction execution

Each discovered vulnerability is documented with a description, precise severity rating and secure-implementation recommendation. Where exploitability is not obvious we include an executable proof-of-concept.

4. Detailed Fix Review and Validation

After the team's remediation we review each fix individually, confirming that corrections eliminate the security risk, do not introduce regressions or new vulnerabilities, and align with EVM best practices. Original proof-of-concept tests are re-run against the fixed code.

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